

Implementing **Log Shipping** and **Database Mirroring** in SQL Server provides strong **Disaster Recovery (DR)** and **High Availability (HA)** options with **minimal downtime** — especially for **critical applications**.

Below is a **comprehensive, step-by-step guide** to implement both, including best practices for failover, performance, and monitoring.

1. Log Shipping in SQL Server

Purpose: Disaster recovery

Edition support: Works with **Standard** and **Enterprise** Editions

Failover type: **Manual**

Downtime: Moderate, but recoverable

Log Shipping Architecture

- **Primary Server:** Main production database
- **Secondary Server(s):** Standby server(s) where logs are applied
- **Jobs:**
 - Backup logs on primary
 - Copy logs to secondary
 - Restore logs on secondary with NORECOVERY or STANDBY

Pre-Requisites

- Databases must be in **FULL recovery mode**
- SQL Server Agent must be running on both servers
- Shared folder or network path for log shipping files

Steps to Implement Log Shipping

Step 1: Set up the Primary Database

-- Ensure FULL recovery

```
ALTER DATABASE [YourDB] SET RECOVERY FULL;
```

-- Take a full backup

```
BACKUP DATABASE [YourDB]  
TO DISK = 'C:\Backups\YourDB_full.bak'  
WITH INIT, FORMAT;
```

-- Take a log backup

```
BACKUP LOG [YourDB]  
TO DISK = 'C:\Backups\YourDB_log.trn';
```

Step 2: Restore on Secondary Server

-- Restore full backup WITH NORECOVERY

```
RESTORE DATABASE [YourDB]  
FROM DISK = '\\PrimaryServer\Backups\YourDB_full.bak'  
WITH NORECOVERY;
```

```
-- Optionally restore log
RESTORE LOG [YourDB]
FROM DISK = '\\PrimaryServer\Backups\YourDB_log.trn'
WITH NORECOVERY;
```

Step 3: Configure Log Shipping via SSMS Wizard

1. On **Primary Server**:
 - Right-click database > **Tasks > Ship Transaction Logs**
 - Check **Enable this as a primary database in log shipping**
 - Configure:
 - Backup settings (folder, schedule)
 - Copy & restore settings on secondary
 - Alert threshold
2. On **Secondary Server**:
 - Add secondary server
 - Choose restore mode: NORECOVERY or STANDBY
 - Set restore schedule

Step 4: Monitor Log Shipping

- Use SSMS > Log Shipping Report
- Query msdb.dbo.log_shipping_monitor_error_detail
- Set alerts for failure on log shipping jobs

Manual Failover Steps (During Disaster)

1. Disable log shipping jobs
2. Restore remaining logs on secondary
3. Recover database:

```
RESTORE DATABASE [YourDB] WITH RECOVERY;
```

4. Re-point applications to secondary

2. Database Mirroring (Deprecated but still used)

Purpose: High availability with automatic or manual failover

Edition support: Standard (manual failover), Enterprise (automatic)

Failover type: Automatic (with witness) or Manual

Downtime: Minimal to zero

Mirroring Roles

- **Principal Server:** Active SQL Server instance
- **Mirror Server:** Secondary server with a copy
- **Witness Server** (optional): Enables automatic failover (Enterprise Edition only)

✅ Pre-Requisites

- Same SQL Server version on all servers
- FULL recovery model on the database
- Use TCP port (default 5022) open on all instances

🚀 Steps to Implement Database Mirroring

📌 Step 1: Backup and Restore

On Principal:

```
BACKUP DATABASE [YourDB]
```

```
TO DISK = 'C:\Backups\YourDB.bak';
```

```
BACKUP LOG [YourDB]
```

```
TO DISK = 'C:\Backups\YourDB_log.trn';
```

On Mirror:

```
RESTORE DATABASE [YourDB]
```

```
FROM DISK = '\\PrimaryServer\Backups\YourDB.bak'
```

```
WITH NORECOVERY;
```

```
RESTORE LOG [YourDB]
```

```
FROM DISK = '\\PrimaryServer\Backups\YourDB_log.trn'
```

```
WITH NORECOVERY;
```

📌 Step 2: Configure Endpoints

-- On both servers

```
CREATE ENDPOINT Mirroring
```

```
STATE = STARTED
```

```
AS TCP (LISTENER_PORT = 5022)
```

```
FOR DATABASE_MIRRORING (
```

```
    ROLE = PARTNER
```

```
);
```

On Witness (if using automatic failover):

```
CREATE ENDPOINT Witness
```

```
STATE = STARTED
```

```
AS TCP (LISTENER_PORT = 5023)
```

```
FOR DATABASE_MIRRORING (ROLE = WITNESS);
```

📌 Step 3: Start Mirroring

On Principal:

```
ALTER DATABASE [YourDB]
```

```
SET PARTNER = 'TCP://MirrorServer:5022';
```

On Mirror:

```
ALTER DATABASE [YourDB]
```

```
SET PARTNER = 'TCP://PrincipalServer:5022';
```

On Principal (to add witness):

```
ALTER DATABASE [YourDB]
```

```
SET WITNESS = 'TCP://WitnessServer:5023';
```

Step 4: Monitor Mirroring

Use SSMS or query:

```
SELECT
    database_id,
    mirroring_state_desc,
    mirroring_role_desc,
    mirroring_safety_level_desc
FROM sys.database_mirroring;
```

Manual Failover (Standard Edition or Maintenance)

ALTER DATABASE [YourDB] SET PARTNER FAILOVER;

 In Standard Edition, automatic failover is **not supported**, only manual.

Comparison: Log Shipping vs Mirroring

Feature	Log Shipping	Database Mirroring
Failover Type	Manual	Manual / Automatic (EE)
Sync Method	Backup/copy/restore	Real-time sync (mirror)
HA or DR	DR	HA
Readable Secondary	No (unless in standby)	No (mirror is inaccessible)
Edition Support	Std/Ent	Std (manual) / Ent (auto)
Latency	Few minutes	Near-zero (synchronous)
Setup Complexity	Moderate	Moderate
Use Case	DR across data centers	HA within region/datacenter

Summary Best Practices

Area	Recommendation
Recovery Model	Always FULL
Network	Low-latency for mirroring; secured shares for log shipping
Monitoring	Set alerts, use jobs to track failures
Performance	Schedule during off-peak for log shipping
Security	Use encrypted endpoints (mirroring)
Automation	Backup job notifications and alerts